

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2020-2021

MATHEMATICS FOR COMPUTER SCIENTISTS (COMP1017)

Time allowed TWO hours only

Candidates must NOT start writing their answers until told to do so

There are FOUR questions in the exam paper. Answer all FOUR questions.

This is an OPEN-BOOK exam. Students may use available resources including Internet, textbooks, and lecture slides to help construct their solutions.

Students are not allowed to communicate with each other through whatsoever means during the examination. Any such communication will be deemed as collusion and may be subject to Academic Misconduct discipline according to Quality Manual.

DO NOT turn examination paper over until instructed to do so

Model Solutions

K – Knowledge
C – Comprehension
A – Application

1. Questions on Sets and Functions (20 marks)

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(i) $\{0,2,12,36,80\} = \{y \in \mathbb{N} \mid \exists x \in \mathbb{N}((y = x^2 + x^3) \wedge (0 \leq x \leq 5))\}.$

*Answers: (K,C)**false*

(ii) Floor is a function from integers to reals.

*Answers: (K)**false*

(iii) $|\{\{1,2,5\}, \{1,2,5\}, 9, \{\}\}| = 4.$

*Answers: (K,C)**false*

(iv) $\emptyset \subseteq \emptyset.$

*Answers: (K,C)**true*

(v) $|2^S| \geq |S|.$

*Answers: (K,C)**true**1 mark for each correct answer.*

(b) Consider the following design of an abstract memory addressing system. You have a multidimensional array of five dimensions. Assume that along each dimension you have 10 elements. The addressing on a specific location is thus $(a_1, a_2, a_3, a_4, a_5)$. The address for the first location is $(0,0,0,0,0)$ while that for the last location is $(n,n,n,n,n), n > 0$. First, construct the complete addresses as a generalized Cartesian product of sets. Second, provide a brief argument why each address generated will be distinct. Third, calculate the size of this array (i.e. how many addresses in total) from your construction. Also show how much of an increase in the array size when each dimension is increased equally from n to $45n$. Full marks for complete construction using basic concepts and definitions in set theory.

Answers: (K,C,A)

First, let $i = 1, 2, 3, 4, 5$, $A_i = \mathbb{N}_0^n = \{0, 1, 2, \dots, n\}$. The complete addressing of the five-dimensional array above can be written in the form of a generalized Cartesian product, using standard set builder notation, as

$$\begin{aligned} A_1 \times A_2 \times A_3 \times A_4 \times A_5 \\ = \{(a_1, a_2, a_3, a_4, a_5) \mid a_1 \in A_1 \wedge a_2 \in A_2 \wedge a_3 \in A_3 \wedge a_4 \in A_4 \wedge a_5 \in A_5\} \end{aligned}$$

[4 marks]

Second, each address is the form of a 5-tuple $(a_1, a_2, a_3, a_4, a_5)$. Each address can be obtained by way of enumeration of an n -ary number system of five-symbol sequence as $(0, 0, 0, 0, 0), (0, 0, 0, 0, 1), \dots, (0, 0, 0, 0, n), (0, 0, 0, 1, 0), (0, 0, 0, 1, 1), \dots, (n, n, n, n, n - 1), (n, n, n, n, n)$.

[3 marks]

Third, since $|A_i| = |\mathbb{N}_0^n| = n$, then the size of the array

$$|A_1 \times A_2 \times A_3 \times A_4 \times A_5| = |A_1| \times |A_2| \times |A_3| \times |A_4| \times |A_5| = n^5.$$

So, if $|B_i| = 45n = 45|A_i|$, then

$$|B_1 \times B_2 \times B_3 \times B_4 \times B_5| = (45n)^5 = 45^5 n^5 = 45^5 |A_1 \times A_2 \times A_3 \times A_4 \times A_5|.$$

[3 marks]

- (c) In your basic construction above, the addresses are elements of a set, which by basic definition has no ordering. Briefly discuss how these addresses can be positioned, which requires imposing some ordering. Higher marks for more formal construction using basic concepts. Hint: consider the notion of a Cartesian coordinate system to represent the position, e.g., $x = (x_1, x_2)$ and Manhattan distance $d(x, y) = |x_1 - y_1| + |x_2 - y_2|$.

[5 marks]

Answers: (K,C,A)

Let $i = 1, 2, 3, 4, 5$. One can endow the original set A_i with the ordering $A_i = (0, 1, 2, \dots, n)$ and distance $d(x_i, y_i) = |x_i - y_i|$.

Then, one can construct a five-dimensional discrete space endowed with the coordinates specified by $A_1 \times A_2 \times A_3 \times A_4 \times A_5$. The origin is set at $(0, 0, 0, 0, 0)$. For each of the five

principal axes (orthogonal to each other), the coordinate is laid out as $(0,1,2, \dots, n)$. Each memory address $\mathbf{a} = (a_1, a_2, a_3, a_4, a_5) \in \mathbf{A}_1 \times \mathbf{A}_2 \times \mathbf{A}_3 \times \mathbf{A}_4 \times \mathbf{A}_5$ is positioned at the corresponding coordinates given by the five-tuple with distances between any pair of points under this coordinate system given by $d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^5 d(x_i, y_i)$.

[5 marks]

2. Questions on Induction and Proof (30 marks)

- (a) Consider this cake cutting and coloring problem. The cake is a perfect circular object. Every new cut is made so that it cuts through the center of the cake and at different angular position. For example, if one cuts a cake horizontally at 0 degrees, then no future cuts are made at the same position. Prove using an induction approach that the cake with finite number of cuts can always be colored black and white alternately. First, specify how the number of cuts is represented and what is your induction hypothesis. Second, construct your proof by induction. For this proof approach, full marks are given for answers further supported by way of graphical illustration.

[18 marks]

Answers: (C,A)

Let $n \in \mathbb{N}$ be the number of cuts.

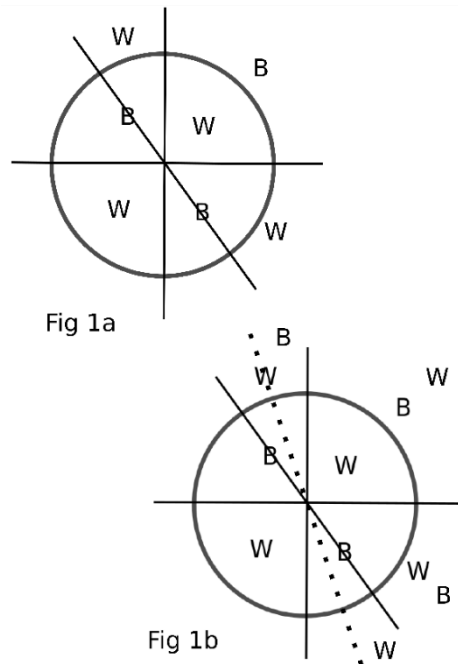
The induction hypothesis is that when a new cut is made, the cake split to two regions by that cut, i.e. one half in the counterclockwise direction (CCW) to the cut, the other half in the clockwise direction (CW). One can just by default let what is in CCW as before. But for CW, one inverts the color.

[3 marks]

Step one (basis, $n = 0$): By definition, where there is no cut, the cake can be colored black or white.

Step two (induction step): Assume that for n cuts, the cake portions are already colored black and white alternately. There are two possibilities when an additional cut is made so that one has $n + 1$ cuts. For both possibilities, the additional cut can be made, and the half region in CW can have its portions' colors inverted (as shown in the figures below).

[5 marks]



[10 marks]

- (b) For the problem above, instead of using graphical illustration, conjecture a formula that can measure the size of the problem. Then use a proof by induction to verify your conjecture for the induction hypothesis. Full marks for complete proof.

[8 marks]

Answers: (K,C,A)

Let $n \in \mathbb{N}$ be the number of cuts.

The conjecture is that the number of portions is given by $S.n = 2n$. The proposition for which one constructs the induction hypothesis is that $P.n \equiv \text{even}(S.n) \equiv \text{true}$. With an even-numbered portions, one can always colored them black and white alternately.

[3 marks]

Step one (basis, $n = 0$): By definition, where there is no cut, the cake has only $S.0 = 2^0 = 1$ portion, which can be colored black or white.

Step two (induction step): By construction,

$$P.n + 1 \equiv \text{even}(S.n + 1)$$

$$= \{\text{definition of } S\}$$

$$\text{even}(2(n + 1))$$

= {simplification}
 even($2n + 2$)
 = {definition of even number}
 true.

[5 marks]

(c) Briefly discuss what differentiates the two proof approaches used.

[4 marks]

Answers: (C)

In the second approach, there is no need to construct that for the additional cut, one can make changes to the coloring so that at the end the portions are colored black and white alternately. The abstraction by way of conjecturing that the number of portions will always be an even number ensures such alternate coloring. However, the first approach provides insight as to how the new cut introduces two new portions.

[4 marks]

3. Questions on Counting and Probabilities.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(K, C)

- (i) There are 125 different ways of lining up 3 people out of 5.
 (false)
- (ii) Choosing 8 students out of 100 students to form a committee is a combination problem.
 (true)
- (iii) If there are 30 students in a class, then at least two have last names that begin with the same letter.
 (true)
- (iv) Throw an unbiased coin 5 times and the probability of obtaining 3 heads is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$.
 (false)
- (v) Put 5 apples into 2 boxes. The probability of one of the boxes having exactly 2 apples is 1.
 (false)

(b) There are 400 students in a programming class. On the midterm test each student received one of the following grades: A, B, C , or D .

- (i) There are less students born in the first half of a year than in the latter. 60% of the grade A students were born in the first half of a year. Show that the event that a randomly chosen student was born in the first half of a year is not independent of the event that a randomly chosen student obtained grade A.

[4 marks]

(C)

Let E be the event that a randomly chosen student was born in the first half of a year.

Let F be the event that a randomly chosen student obtained grade A.

We have that $P(E) < 0.5$ [1 mark]

$P(E|F) = 0.6$ [1 mark]

To check if E and F are independent of each other, we would like to see if $P(E) = P(E|F)$ [1 marks]

Clearly, $P(E) < 0.5 < 0.6 = P(E|F)$, therefore $P(E) \neq P(E|F)$, E and F are not independent. [1 mark]

- (ii) It turned out that the number of students receiving the grade A, B, C, and D are 95, 120, 110, 75, respectively. What is the probability of 10 randomly chosen students having 3 A's, 3 B's, 2 C's, and 2 D's? Explain.

[6 marks]

(C, A)

There are in total $C(400, 10)$ ways of choosing 10 students out of 400 students. [1 mark]

There are $C(95, 3)$ ways of choosing 3 students out of 95 students who have obtained grade A. [1 mark]

There are $C(120, 3)$ ways of choosing 3 students out of 120 students who have obtained grade B. [1 mark]

There are $C(110, 2)$ ways of choosing 2 students out of 110 students who have obtained grade C. [1 mark]

There are $C(75, 2)$ ways of choosing 2 students out of 75 students who have obtained grade D. [1 mark]

The probability is then $(C(95, 3) * C(120, 3) * C(110, 2) * C(75, 2)) / C(400, 10) =$

0.00027546447. [1 mark]

(c) A Machine Learning algorithm was developed to predict whether an input image contains a human face or not. It was tested on 800 independently randomly selected images and the prediction errors were recorded. An error occurs when an input image does not contain a human face and the algorithm predicts otherwise, or when an input image contains a human face and the algorithm predicts otherwise.

- (i) Identify all the relevant random variables and their respective sample spaces in the scenario given above.

[2 marks]

(K, C)

Input image X: $\{1, 0\}$ where $X=1$ means the input image X contains a human face, $X=0$ means the input image X does not contain a human face. [1 mark, 0.5 for identifying X, 0.5 for sample space $\{1,0\}$]

Prediction Y: $\{1, 0\}$ where $Y=1$ mean the prediction of containing human face, $Y=0$ means the prediction of containing no human face. [1 mark, 0.5 for identifying Y, 0.5 for sample space $\{1,0\}$]

- (ii) Express the prediction error as a random event and provide a formula to calculate its probability based on the frequentist's definition of probability.

[3 marks]

(C, A)

Let E denote the prediction error. It is a result of running experiment of predicting the output Y for an input image X by observing the input value and output value jointly, i.e., (X,Y) . [1 mark if the rest is not provided]

The sample space of (X,Y) is $\{(1,1), (1,0), (0,1), (0,0)\}$. [1 mark]

Thus $E=\{(1,0), (0,1)\}$. [1 mark]

$P(E)=(\#(1,0) + \#(0,1))/800$, where #a denotes the number of a. [1 mark]

- (iii) Out of the 800 testing images, 30% contain a human face for which 20% were predicted wrongly. On the other hand, 65% of the images that contain no human faces were predicted correctly. Now the algorithm predicts for a new unseen image that it contains a human face. What is the probability that the new unseen image indeed contains a human face?

[5 marks]

(C, P)

Using Bayes rule, we have

$$\begin{aligned}
 P(X=1|Y=1) &= (P(Y=1|X=1) * P(X=1)) / P(Y=1) \\
 &= (P(Y=1|X=1) * P(X=1)) / ((P(Y=1|X=1) * P(X=1) + (P(Y=1|X=0) * \\
 &P(X=0)) \text{ [1 mark]}
 \end{aligned}$$

$$P(X=0) = 0.7, P(X=1) = 0.3 \text{ [1 mark]}$$

$$P(Y=0|X=1) = 0.2, P(Y=1|X=1) = 0.8 \text{ [1 mark]}$$

$$P(Y=0|X=0) = 0.65, P(Y=1|X=0) = 0.35 \text{ [1 mark]}$$

Substituting, we have

$$P(X=1|Y=1) = (0.8 * 0.3) / (0.8*0.3 + 0.35*0.7) = 0.49484536082 \text{ [1 mark]}$$

4. Questions on Numbers and Relations.

(a) For each statement, say whether it is **true** or **false**.

[5 marks]

(K, C)

(i) $\lfloor \lceil x \rceil \rfloor = \lfloor x \rfloor$ for all real numbers x .

(true)

(ii) -67 is congruent to 3 modulo 7 .

(true)

(iii) $\gcd(100, 104) = 2$.

(false)

(iv) If $a|bc$, where a, b , and c are positive integers and $a \neq 0$, then $a|b$ or $a|c$.

(false)

(v) For integers $a, b, m \neq 0$, $a \equiv b \pmod{m}$ is a reflexive relation between a and b .

(true)

(b) Let $\emptyset$ be an empty set. Show that the relation $R = \emptyset$ on a nonempty set S is symmetric and transitive, but not reflexive.

[6 marks]

(K, C)

To prove that R is symmetric on S , we need to show that $\forall a, b \in S, (a, b) \in R \rightarrow (b, a) \in R$ [1 mark]. However, since R is empty, then $\forall a, b \in S, (a, b) \in R$ is false, hence $(a, b) \in R \rightarrow (b, a) \in R$ is true, i.e., symmetric. [1 mark]

To prove that R is transitive, we need to show that $\forall a, b, c \in S, (a, b) \wedge (b, c) \in R \rightarrow (a, c) \in R$ [1 mark]. However, since R is empty, then $\forall a, b, c \in S, (a, b) \wedge (b, c) \in R$ is false, hence $(a, b) \wedge (b, c) \in R \rightarrow (a, c) \in R$ is true, i.e., transitive. [1 mark]

To prove that R is reflexive, we need to show that $\forall a \in S, (a, a) \in R$ [1 mark]. However, since R is empty, $(a, a) \in R$ is false for all $a \in S$. Thus, R is not reflexive. [1 mark]

(c) Let R be the relation $\{(1, 2), (1, 3), (2, 3), (2, 4), (3, 1)\}$, and let S be the relation $\{(2, 1), (3, 1), (3, 2), (4, 2)\}$. Find $S \circ R$.

[4 marks]

(K, C)

$\{(1, 1), (1, 2), (2, 1), (2, 2)\}$ [1 mark each]

(d) How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \dots, 1000\}$ consisting of the first 1000 positive integers have if R is

- (i) $\{(a, b) \mid a \leq b\}$?
- (ii) $\{(a, b) \mid a = b \pm 1\}$?
- (iii) $\{(a, b) \mid a + b = 1000\}$?
- (iv) $\{(a, b) \mid a + b \leq 1001\}$?
- (v) $\{(a, b) \mid a \neq 0\}$?

Sketch the matrix for each of the above relations.

[10 marks]

(K, C)

(i) $1000+999+\dots+2+1 = 1000*1001/2 = 500500$ [2 marks]

(ii) $2*999 = 1998$ [2 marks]

(iii) 999 [2 marks]

(iv) $1000*1000-(i) = 499500$ [2 marks]

(v) $1000*1000 = 10^6$ [2 marks]

For each answer, 1 mark for the correct sketch of the matrix and 1 mark for the correct answer.